SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1025	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 1 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Problem Statement:	**Bus Ticket Reservation System**		
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Date:	15-10-2026	**Duration:**	60 minutes

Code of Conduct

I K Sampath Reddy (192424285) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *__K Sampath Reddy____*

Annexure A – Scenario-Based Requirement Analysis

BusBuddy: A Smart Bus Ticket Reservation System

Scenario-Based Requirement Analysis and Software Requirement Specification (SRS)

No.	Section (as required in Annexure A)	Key content
1	Study and Analyze the Scenario	Scenario, domain, stakeholders, challenges, objectives
2	Identify Functional and Non-Functional Requirements	47 functional and 22 non-functional requirements
3	Identify Actors and Use Cases	8 actors, 23 use cases, UML use case diagram
4	Prepare the Software Requirement Specification (SRS)	Introduction to system features (9 parts)
5	Identify Assumptions and Constraints	Assumptions; technical, operational, economic, legal, time constraints
6	Validate Requirement Completeness and Consistency	Coverage, quality checks, improvements

1. Study and Analyze the Scenario

1.1 The Scenario

Requirements are best understood through a real situation. The scenario below shows what happens today in a typical bus service, and what BusBuddy must change.

> **Scenario: Meera's journey**
> It is Friday evening in Chennai. **Meera** needs to reach Bengaluru early on Saturday. She phones an agent, who says seat 12 on the 10 PM bus is free. At the counter, the clerk has just sold seat 12 to someone else, so Meera is offered a seat in the last row. She pays cash, receives a handwritten ticket and is told refunds need a visit to the counter.
>
> Meanwhile, the **operator's manager** cannot tell how many seats on that bus are really empty, the **conductor** has no reliable passenger list, and the bus leaves 20 minutes late with nobody informing the passengers.

In my own words: the real problem is not showing a list of buses. It is giving every user the **same, correct, real-time view of seats**, making sure **one seat is sold only once**, and handling payment, tickets and refunds without manual work. The table links each event in the scenario to what the system must do.

What happened in the scenario	What the system must do (requirements)
Seat 12 was sold twice (agent and counter disagree).	Show a live seat map and lock a seat while one user pays (FR-12 to FR-16).
Meera could only pay cash and got a handwritten ticket.	Accept online payment and issue an e-ticket with a QR code (FR-22 to FR-26).
A refund needs a visit to the counter.	Cancel online and calculate the refund automatically (FR-28 to FR-31).
The manager does not know the real seat occupancy.	Provide a dashboard with sales and occupancy (FR-41, FR-42).
The conductor has no reliable passenger list.	Let the conductor scan QR tickets and update trip status (FR-43, FR-44).
The bus left late and nobody was informed.	Send delay alerts and show live bus location (FR-32, FR-33).

1.2 Business Domain

BusBuddy belongs to the **passenger road-transport and online ticketing** domain (often called travel technology). Its business is selling a limited, time-bound resource (a seat on one trip) to many users at the

same moment. This makes **availability, timing and trust** the key concerns. The main domain terms used in this document are:

Term	Meaning in BusBuddy
Route	A journey between a source and a destination, with ordered boarding points.
Trip	One scheduled run of a bus on a route on a given date and time, with its own fare.
Seat layout	The arrangement of seats in a bus (seater or sleeper, for example 2+2 or 2+1).
Boarding point	A stop on the route where a passenger may board.
Seat lock	A short hold (8 minutes) on selected seats while the passenger completes payment.
Refund slab	A time-based rule that decides how much of the fare is refunded on cancellation.
Waitlist	A queue of passengers who want a seat on a full trip.
e-Ticket / QR code	A digital ticket with a unique code that the conductor scans at boarding.

1.3 Stakeholders

Stakeholder	Description	Key needs
Passengers	Travellers who search, book, pay, travel and cancel.	Real seat availability, fast booking, safe payment, fair refunds, safety, trip updates.
Bus Operator / Admin	Owns buses and manages routes, trips, fares and reports.	Control over schedules and fares, no double booking, live sales and occupancy data.
Conductors / Drivers	Verify tickets and update trip status.	Quick QR scanning, accurate passenger list, easy status update.
Payment Gateway Provider	External company that processes payments and refunds.	Correct integration, secure data handling, compliance with payment rules.
SMS / Email and Maps Providers	External services for alerts and live location.	Valid API use within usage limits.
Transport Authority	Regulates fares, safety and records.	Accurate, tamper-proof records and reports.
Customer Support Team	Handles complaints and booking queries.	Tools to find a booking and fix problems quickly.
Development and Maintenance Team	Builds, tests and maintains the system.	Clear, testable, stable requirements.

1.4 Existing Challenges

Challenge	Effect	Requirements that address it
Seat information is outdated or not shared	Double booking, wrong information, lost sales	FR-12, FR-14 to FR-16, NFR-02, NFR-08
Manual, cash-only payment	Long queues, risk of mistakes and theft	FR-22 to FR-24, NFR-05
Paper tickets	Forgery, loss, slow checking	FR-25, FR-43
Manual cancellation and refund	Delays and disputes	FR-28 to FR-31, FR-40
No sales or occupancy data	Poor planning of buses and fares	FR-41, FR-42
Passengers are not informed of delays	Missed buses, complaints	FR-32 to FR-35
Safety concerns, especially for women travelling alone	Reduced trust in the service	FR-17, FR-19, FR-32

1.5 System Objectives

ID	Objective	Measure of success
O-1	Fast, simple booking	Median time from search to confirmed ticket of 3 minutes or less (NFR-13).
O-2	Zero double booking	No seat sold twice (NFR-08).
O-3	Secure online payment and automatic refund	At least 98% payment success; refund amount shown instantly (FR-22 to FR-31).
O-4	Paperless, verifiable tickets	100% of tickets carry a unique QR code verified by the conductor (FR-25, FR-43).
O-5	Operator control and insight	Admin can manage buses, routes, trips and see live reports (FR-36 to FR-42).
O-6	Safe, reliable and scalable service	99.5% uptime, support for 5,000 concurrent users (NFR-11, NFR-16).

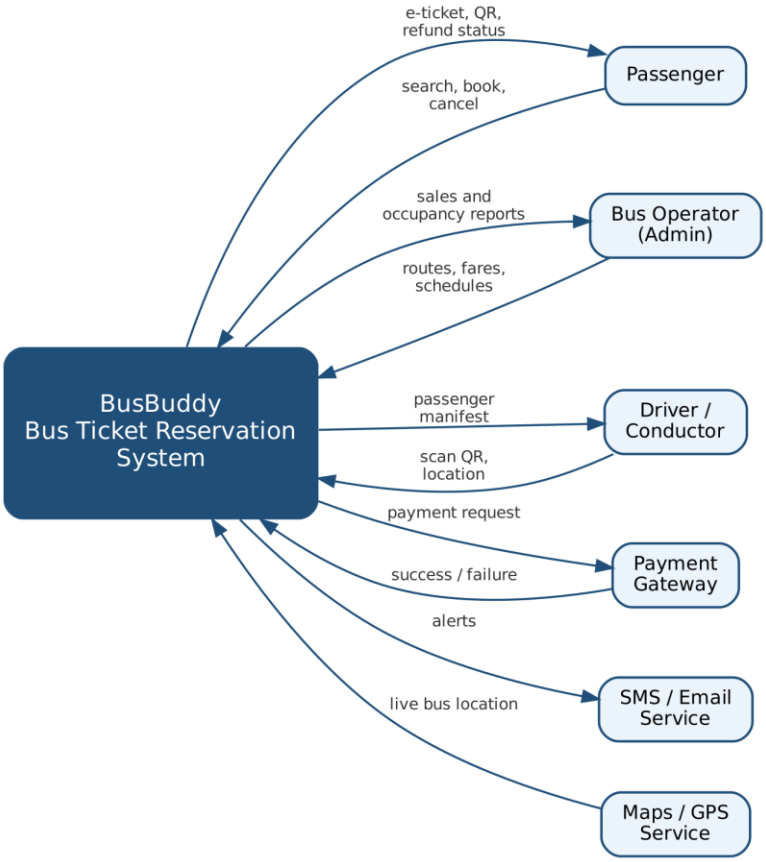


Figure 1: *Context diagram showing how BusBuddy interacts with users and external services*

2. Identify Functional and Non-Functional Requirements

2.1 How the Requirements Were Identified

Requirements were derived from four sources: the scenario walk-through (Section 1.1), the needs of each stakeholder (Section 1.3), the weaknesses of the current process (Section 1.4) and the legal and technical rules that apply (Section 5). Each requirement has a unique ID, a priority and a responsible actor, so it can be traced to a use case and tested.

Priority	Meaning (MoSCoW)	Rule of thumb
Must	Must have	The system cannot be released or is unusable without it.
Should	Should have	Important and expected, but a short-term workaround exists.
Could	Could have	Useful extra; included only if time and budget allow.

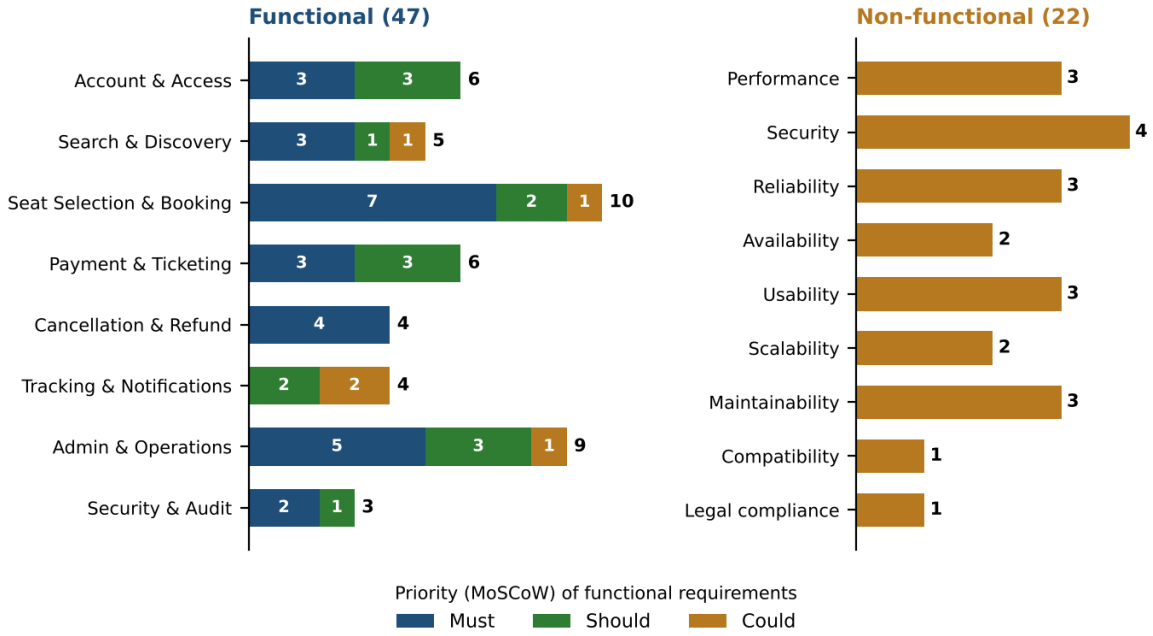


Figure 2: *Requirements by module and category (47 functional, 22 non-functional)*

2.2 Functional Requirements

The functional requirements describe what the system shall do. There are 47 requirements in eight modules: 27 Must, 15 Should and 5 Could.

ID	Requirement	Priority	Actor
1. Account & Access			
FR-01	The system shall allow a guest to register using name, email, mobile number and password.	Must	Guest
FR-02	The system shall verify the mobile number or email with a 6-digit OTP that expires in 5 minutes.	Must	Guest
FR-03	The system shall allow registered users to log in using a password or an OTP.	Must	Passenger
FR-04	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.	Should	Passenger
FR-05	The system shall allow users to reset a forgotten password using an OTP.	Should	Passenger
FR-06	The system shall allow users to view and update their profile details.	Should	Passenger
2. Search & Discovery			
FR-07	The system shall allow users to search buses by source, destination and travel date.	Must	Guest
FR-08	The system shall show, for each result, the bus type, departure and arrival time, duration, fare and seats left.	Must	Guest
FR-09	The system shall let users filter and sort results by price, departure time and bus type.	Should	Guest
FR-10	The system shall list only future trips, and mark trips with no free seats as Full.	Must	Guest
FR-11	The system shall suggest nearby dates when no bus is available on the chosen date.	Could	Guest
3. Seat Selection & Booking			
FR-12	The system shall display a live seat map showing available, locked, booked and ladies seats.	Must	Passenger
FR-13	The system shall allow selection of up to 6 seats in one booking.	Must	Passenger
FR-14	The system shall lock the selected seats for 8 minutes and show a countdown timer.	Must	Passenger
FR-15	The system shall release locked seats automatically on timeout, payment failure or user cancellation.	Must	System
FR-16	The system shall never confirm the same seat of a trip for two different bookings.	Must	System
FR-17	The system shall apply the women-safe seat rule: an unrelated male passenger is not placed next to a ladies seat.	Should	Passenger

ID	Requirement	Priority	Actor
FR-18	The system shall capture name, age and gender for every traveller in a booking.	Must	Passenger
FR-19	The system shall let the passenger choose a boarding point and a drop point from the route.	Should	Passenger
FR-20	The system shall maintain a waitlist for full trips and confirm the first waitlisted passenger when a seat is released.	Could	Passenger
FR-21	The system shall show a fare summary (base fare, taxes, convenience fee, total) before payment.	Must	Passenger
4. Payment & Ticketing			
FR-22	The system shall accept online payment through the payment gateway (UPI, cards, net banking).	Must	Passenger
FR-23	The system shall confirm a booking only after the payment gateway reports success.	Must	System
FR-24	The system shall allow a failed payment to be retried while the seat lock is active.	Should	Passenger
FR-25	The system shall generate an e-ticket with a booking ID and a unique QR code.	Must	Passenger
FR-26	The system shall send the e-ticket by SMS and email within 1 minute of confirmation.	Should	System
FR-27	The system shall let users download e-tickets as PDF and view their booking history.	Should	Passenger
5. Cancellation & Refund			
FR-28	The system shall allow cancellation of a confirmed ticket before departure.	Must	Passenger
FR-29	The system shall calculate the refund automatically using time slabs configured by the admin.	Must	System
FR-30	The system shall initiate the refund to the original payment method and show its status.	Must	System
FR-31	The system shall make cancelled seats available for sale immediately.	Must	System
6. Tracking & Notifications			
FR-32	The system shall show the live bus location and estimated arrival time to passengers with a booking.	Could	Passenger
FR-33	The system shall send a delay alert when a trip is delayed by more than 15 minutes.	Could	System
FR-34	The system shall send reminders 24 hours and 2 hours before departure.	Should	System
FR-35	The system shall notify the passenger about cancellation and refund updates.	Should	System

ID	Requirement	Priority	Actor
7. Admin & Operations			
FR-36	The system shall restrict admin functions to users with the Admin role.	Must	Admin
FR-37	The system shall allow the admin to add, edit and deactivate buses with their seat layout.	Must	Admin
FR-38	The system shall allow the admin to create and manage routes with boarding points.	Must	Admin
FR-39	The system shall allow the admin to schedule single and recurring trips and set fares by seat type.	Must	Admin
FR-40	The system shall allow the admin to configure refund slabs and the convenience fee.	Should	Admin
FR-41	The system shall provide a dashboard of sales, seat occupancy and cancellations.	Should	Admin
FR-42	The system shall allow reports to be exported as PDF or Excel.	Could	Admin
FR-43	The conductor app shall scan a QR code and report whether the ticket is valid, already used or for another trip.	Must	Conductor
FR-44	The system shall let the conductor update the trip status (started, delayed, completed).	Should	Conductor
8. Security & Audit			
FR-45	The system shall record an audit log of admin actions and payment events.	Must	System
FR-46	The system shall store passwords only as salted hashes and shall never store card details.	Must	System
FR-47	The system shall end idle sessions after 15 minutes for admins and 30 minutes for passengers.	Should	System

2.3 Non-Functional Requirements

Non-functional requirements describe how well the system must work. Every requirement carries a measurable target so that it can be tested. The categories include performance, security, reliability, availability, usability, scalability and maintainability, as asked in the assessment, plus compatibility and legal compliance.

ID	Requirement	Measurable target
1. Performance		
NFR-01	Search results shall load within 3 seconds for 95% of requests with 1,000 concurrent users.	3 s at the 95th percentile
NFR-02	The live seat map shall refresh within 2 seconds of any seat change.	2 s
NFR-03	A booking shall be confirmed within 5 seconds after payment success is received.	5 s

ID	Requirement	Measurable target
2. Security		
NFR-04	All traffic shall be encrypted with TLS 1.2 or higher.	100% of requests over HTTPS
NFR-05	Payments shall go through a PCI-DSS compliant gateway; the system shall not store card data.	0 card numbers stored
NFR-06	OTPs shall expire in 5 minutes, be single use, and be rate-limited to 5 requests per hour per user.	Rate limit enforced
NFR-07	The system shall be protected against the OWASP Top 10 risks (for example SQL injection and cross-site scripting).	0 high-severity findings at release
3. Reliability		
NFR-08	The system shall guarantee zero double booking of a seat through a database uniqueness rule.	0 duplicate seat sales
NFR-09	No payment shall remain without a booking: unmatched payments shall be reconciled or refunded automatically.	Within 15 minutes
NFR-10	Data shall be backed up daily with a recovery point of 1 hour and a recovery time of 4 hours.	RPO 1 h, RTO 4 h
4. Availability		
NFR-11	The system shall be available 99.5% of the time each month, excluding planned maintenance (2 AM to 4 AM).	99.5% uptime
NFR-12	If the SMS service fails, notifications shall be sent by email automatically.	Fallback within 2 minutes
5. Usability		
NFR-13	A first-time user shall be able to complete a booking in 3 minutes or less without help.	3 min in usability tests
NFR-14	The interface shall be mobile-first and fully usable on screens 360 px wide and above.	No horizontal scrolling
NFR-15	The interface shall meet WCAG 2.1 Level AA (labels, contrast, keyboard use).	AA checklist passed
6. Scalability		
NFR-16	The system shall support 5,000 concurrent users by adding servers, without code changes.	5,000 users
NFR-17	The system shall support 500 buses and 100,000 bookings per month without redesign.	500 buses, 100,000 bookings

ID	Requirement	Measurable target
7. Maintainability		
NFR-18	Code shall be modular with at least 80% unit-test coverage on business logic.	80% coverage
NFR-19	Lock time, refund slabs and fees shall be changeable by configuration, without redeployment.	0 code changes
NFR-20	Errors shall be logged centrally and an alert shall be raised within 5 minutes of a critical failure.	5 min
8. Compatibility		
NFR-21	The web app shall work on the latest two versions of Chrome, Edge, Firefox and Safari, and on Android 9+ and iOS 14+.	Test matrix passed
9. Legal compliance		
NFR-22	Personal data shall be collected with consent and deleted or anonymised on request, as required by the Digital Personal Data Protection Act, 2023.	Request handled within 30 days

3. Identify Actors and Use Cases

3.1 Actors

An actor is a person or an external system that interacts with BusBuddy. **Primary actors** start an interaction to achieve a goal. **Secondary actors** support the system and are called by it.

Actor	Type	Description	Main goal
Guest (Visitor)	Primary	A person who is not logged in.	Search buses and register.
Passenger	Primary	A registered, logged-in user (a specialisation of Guest).	Book, pay, travel, cancel and track.
Admin / Operator	Primary	Staff of the bus operator.	Manage buses, routes, trips, fares and reports.
Conductor	Primary	Crew member who boards passengers.	Verify tickets and update trip status.
Payment Gateway	Secondary	External payment system.	Process payments and refunds.
SMS / Email Service	Secondary	External messaging service.	Deliver OTPs, tickets and alerts.
Maps / GPS Service	Secondary	External location service.	Provide bus location and arrival time.
Transport Authority	Secondary	Government regulator.	Receive reports and audit records.

3.2 Use Cases for Each Actor

Actor	Use cases
Guest	UC-01 Register Account; UC-03 Search Buses
Passenger	UC-02 Log In; UC-03 Search Buses; UC-04 Select Seats (includes UC-05 Lock Seats); UC-06 Book Ticket (includes UC-07 Make Payment; extended by UC-08 Join Waitlist); UC-09 View / Download e-Ticket; UC-10 Cancel Ticket (includes UC-11 Calculate and Issue Refund); UC-12 View Booking History; UC-13 Manage Profile; UC-14 Track Bus
Admin / Operator	UC-02 Log In; UC-16 Manage Buses; UC-17 Manage Routes and Boarding Points; UC-18 Schedule Trips and Set Fares; UC-19 Configure Refund Slabs; UC-20 View Sales and Occupancy Reports (extended by UC-21 Export Report)
Conductor	UC-02 Log In; UC-22 Scan QR Ticket; UC-23 Update Trip Status
Payment Gateway	UC-07 Make Payment; UC-11 Calculate and Issue Refund
SMS / Email Service	UC-15 Send Notifications (also used for OTPs in UC-01, UC-02 and UC-13)
Maps / GPS Service	UC-14 Track Bus
Transport Authority	UC-21 Export Report

3.3 Use Case Catalogue

ID	Use case	Primary actor	Secondary actor	Related requirements
UC-01	Register Account	Guest	-	FR-01, 02, 46
UC-02	Log In	Passenger, Admin, Conductor	-	FR-03, 04, 05, 47
UC-03	Search Buses	Guest, Passenger	-	FR-07, 08, 09, 10, 11
UC-04	Select Seats	Passenger	-	FR-12, 13, 17
UC-05	Lock Seats	Passenger (included)	-	FR-14, 15
UC-06	Book Ticket	Passenger	Payment Gateway	FR-16, 18, 19, 21
UC-07	Make Payment	Passenger (included)	Payment Gateway	FR-22, 23, 24
UC-08	Join Waitlist	Passenger (extends UC-06)	-	FR-20
UC-09	View / Download e-Ticket	Passenger	-	FR-25, 27
UC-10	Cancel Ticket	Passenger	-	FR-28, 31
UC-11	Calculate and Issue Refund	Passenger (included)	Payment Gateway	FR-29, 30
UC-12	View Booking History	Passenger	-	FR-27
UC-13	Manage Profile	Passenger	-	FR-06
UC-14	Track Bus	Passenger	Maps / GPS Service	FR-32
UC-15	Send Notifications	System (triggered)	SMS / Email Service	FR-26, 33, 34, 35
UC-16	Manage Buses	Admin	-	FR-36, 37, 45
UC-17	Manage Routes and Boarding Points	Admin	-	FR-38
UC-18	Schedule Trips and Set Fares	Admin	-	FR-39
UC-19	Configure Refund Slabs	Admin	-	FR-40
UC-20	View Sales and Occupancy Reports	Admin	-	FR-41
UC-21	Export Report	Admin (extends UC-20)	Transport Authority	FR-42
UC-22	Scan QR Ticket	Conductor	-	FR-43
UC-23	Update Trip Status	Conductor	-	FR-44

3.4 Use Case Diagram

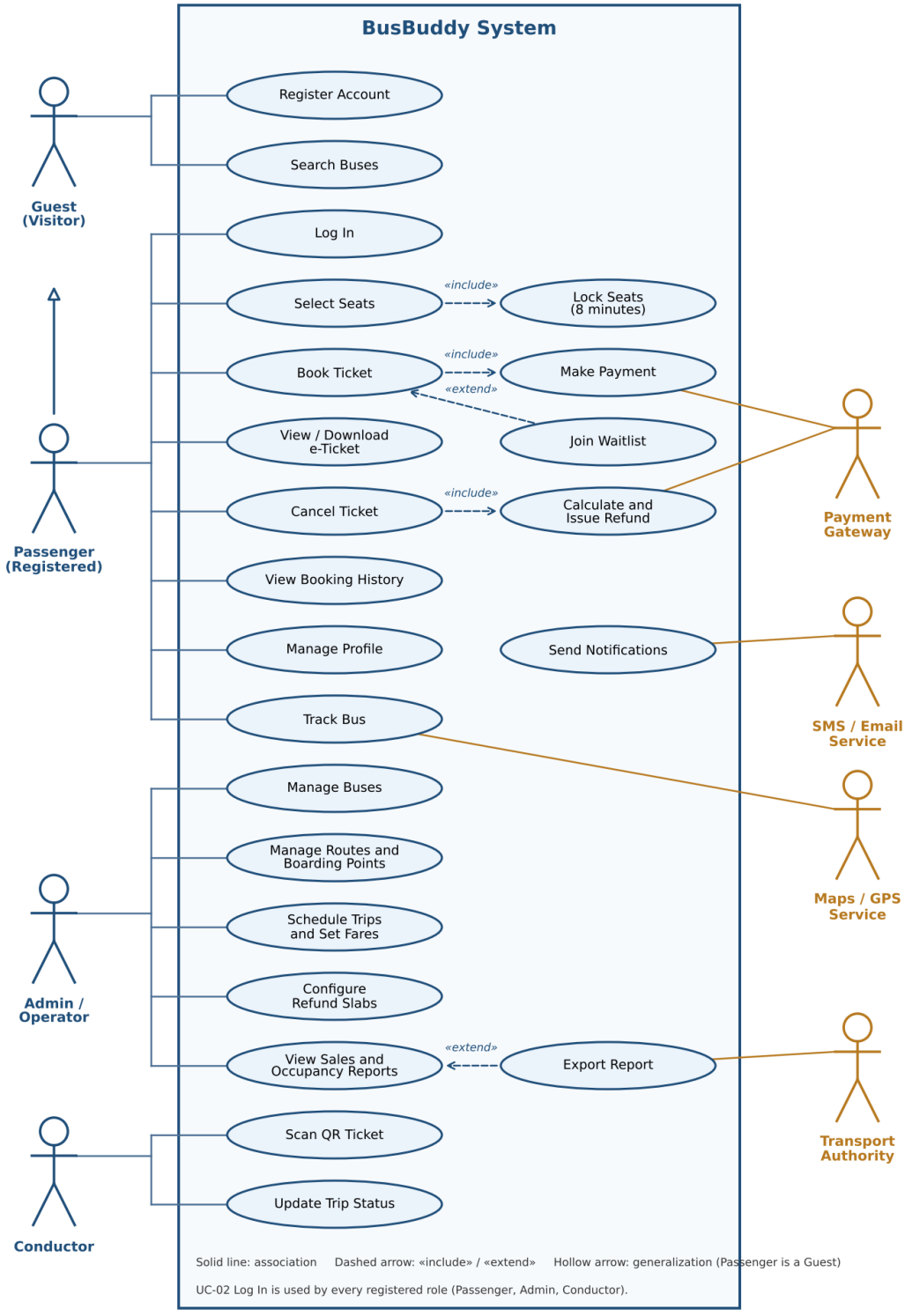


Figure 3: *UML use case diagram of BusBuddy*

Notation: stick figures are actors (blue for primary, gold for secondary); ovals are use cases; the large rectangle is the system boundary. «include» means the base use case always uses the other one; «extend» means the use case adds optional behaviour to the base use case.

4. Prepare the Software Requirement Specification (SRS)

This section is the Software Requirement Specification of BusBuddy. It follows the structure recommended in IEEE 830 / ISO/IEC/IEEE 29148 and contains the nine parts asked for in the assessment.

4.1 Introduction

4.1.1 Purpose

This SRS defines the requirements of the **BusBuddy Bus Ticket Reservation System**, version 1.0. It states what the system must do and how well, so that stakeholders can agree on the product, developers can design and build it, and testers can verify it.

4.1.2 Intended Audience

Reader	Use of this document
Bus operator and Product Owner	Confirm that the needs of the business are captured and prioritised.
Developers and architects	Design and implement the system from the requirements.
Testers (QA)	Write test cases from the requirements and use cases.
Project guide / evaluator	Assess the completeness and quality of the analysis.

4.1.3 Definitions and Abbreviations

Term	Definition
SRS	Software Requirement Specification
FR / NFR	Functional Requirement / Non-Functional Requirement
UC	Use Case
OTP	One-Time Password sent by SMS or email for verification
QR code	Quick Response code printed on the e-ticket and scanned at boarding
UPI	Unified Payments Interface, an Indian instant payment method
PCI-DSS	Payment Card Industry Data Security Standard
DPDP Act	Digital Personal Data Protection Act, 2023 (India)
ETA	Estimated Time of Arrival
RPO / RTO	Recovery Point / Recovery Time Objective (data loss and downtime allowed after a failure)

4.1.4 References

- IEEE Std 830-1998 and ISO/IEC/IEEE 29148:2018, Requirements Engineering.
- Digital Personal Data Protection Act, 2023 (India).
- PCI Data Security Standard (PCI-DSS), PCI Security Standards Council.
- W3C Web Content Accessibility Guidelines (WCAG) 2.1.
- OWASP Top 10 web application security risks.

4.1.5 Document Conventions

Requirements use unique IDs: **FR-nn** (functional), **NFR-nn** (non-functional), **UC-nn** (use case), **DR-nn** (data rule) and **SF-n** (system feature). The word "shall" marks a mandatory statement. Priorities follow MoSCoW (Must, Should, Could).

4.2 Problem Description

Bus operators sell seats through counters, agents and phone calls using paper charts. Seat information is not shared in real time, so the same seat can be sold twice, payment and refunds are manual, and neither the operator nor the conductor has reliable data. Passengers lose time and trust, and operators lose revenue. The problem is to build a system that gives all users one real-time view of seats, lets a passenger complete a booking online without conflicts, and gives the operator and conductor the tools to run trips efficiently.

Core problem	Required capability
Same seat sold to two people	Seat lock and a database rule that prevents duplicate confirmation
Slow, manual booking and payment	Online search, booking and payment in a few minutes
Unreliable tickets and boarding	QR e-ticket verified by the conductor
Manual cancellation and refund	Automatic, rule-based refund
No visibility for the operator	Real-time dashboard and reports

4.3 Scope of the System

In Scope (version 1.0)	Out of Scope
• Registration, login and profile • Bus, route, trip and fare management • Search, live seat map and 8-minute seat lock • Online payment, e-ticket and QR code • Cancellation with automatic refund • Waitlist and women-safe seating • SMS / email alerts, live tracking • Admin reports and conductor QR scanning	• Buying or maintaining buses • Driver payroll and vehicle maintenance • Hotel, cab or train booking • International routes and multi-currency payments • Manufacturing of GPS hardware • Tax filing and permit processing

System boundary: BusBuddy ends where the external services begin. It sends payment requests to the gateway, messages to the SMS / email service and location queries to the maps service, but it does not process cards, deliver messages or own the map data itself.

4.4 Functional Requirements (Summary)

The complete list of functional requirements is given in Section 2.2. The table summarises them by module and priority.

Module	IDs	Total	Must	Should	Could
Account & Access	FR-01 to FR-06	6	3	3	0
Search & Discovery	FR-07 to FR-11	5	3	1	1
Seat Selection & Booking	FR-12 to FR-21	10	7	2	1

Module	IDs	Total	Must	Should	Could
Payment & Ticketing	FR-22 to FR-27	6	3	3	0
Cancellation & Refund	FR-28 to FR-31	4	4	0	0
Tracking & Notifications	FR-32 to FR-35	4	0	2	2
Admin & Operations	FR-36 to FR-44	9	5	3	1
Security & Audit	FR-45 to FR-47	3	2	1	0
Total	**FR-01 to FR-47**	**47**	**27**	**15**	**5**

4.5 Non-Functional Requirements (Summary)

The complete list is given in Section 2.3. The table shows the key target for each category.

Category	IDs	Key target
Performance	NFR-01 to NFR-03	Search in 3 s (95th percentile)
Security	NFR-04 to NFR-07	TLS 1.2+, no card data stored, OWASP Top 10 protected
Reliability	NFR-08 to NFR-10	Zero double booking; RPO 1 h, RTO 4 h
Availability	NFR-11 to NFR-12	99.5% monthly uptime
Usability	NFR-13 to NFR-15	Booking in 3 minutes; WCAG 2.1 AA
Scalability	NFR-16 to NFR-17	5,000 concurrent users; 500 buses
Maintainability	NFR-18 to NFR-20	80% unit-test coverage; configuration without redeployment
Compatibility	NFR-21	Latest 2 versions of major browsers; Android 9+, iOS 14+
Legal compliance	NFR-22	DPDP Act, 2023 consent and deletion

4.6 Use Case Descriptions

Eight key use cases are described in detail below. They cover registration, search, seat selection, booking, waitlist, cancellation, trip scheduling and ticket scanning, which together carry the most important business rules.

UC-01 · Register Account	
Primary actor	Guest
Secondary actors	SMS / Email Service (OTP delivery through UC-15)
Description	A visitor creates a BusBuddy account so that tickets can be booked.
Preconditions	The visitor is not logged in and has a mobile number or email address.
Trigger	The guest selects Sign up.
Main flow	1. The guest opens the registration page. 2. The guest enters name, email, mobile number and password. 3. The system validates the format and checks that the email and mobile number are unique.

	4. The system creates a pending account and sends a 6-digit OTP. 5. The guest enters the OTP. 6. The system verifies the OTP, marks the account Verified and opens the home page.
Alternate / exception flows	3a. Email or mobile already registered: the system shows "Already registered" and offers Log In. 3b. Weak password: an inline message explains the password rules. 5a. Wrong OTP three times: the OTP is blocked and a new one must be requested. 5b. OTP older than 5 minutes: the system rejects it and offers to resend.
Postconditions	A verified account exists; the password is stored as a salted hash.
Related requirements	FR-01, FR-02, FR-46, NFR-06

UC-03 · Search Buses	
Primary actor	Guest, Passenger
Secondary actors	None
Description	The user searches for buses on a route and date.
Preconditions	At least one trip has been scheduled by the admin.
Trigger	The user enters a search.
Main flow	1. The user enters source, destination and travel date. 2. The system checks that source and destination differ and the date is not in the past. 3. The system lists matching trips with bus type, departure and arrival time, fare and seats left. 4. The user filters or sorts the results. 5. The user selects a trip and moves to UC-04 Select Seats.
Alternate / exception flows	2a. Invalid input: the system shows a clear error message. 3a. No trip on that date: the system suggests nearby dates. 3b. Trip has no free seats: the system shows Full and offers UC-08 Join Waitlist.
Postconditions	Results are displayed; no data is changed.
Related requirements	FR-07 to FR-11, NFR-01

UC-04 · Select Seats	
Primary actor	Passenger
Secondary actors	None (includes UC-05 Lock Seats)
Description	The passenger chooses seats on a live seat map and the system locks them for 8 minutes.
Preconditions	The passenger is logged in and has selected a trip that has free seats.
Trigger	The passenger opens the seat map.
Main flow	1. The system shows the live seat map (available, booked, locked and ladies seats).

	2. The passenger selects 1 to 6 seats. 3. The system applies the women-safe seat rule. 4. The system locks the seats for 8 minutes (UC-05) and shows a countdown. 5. The system shows the fare summary and continues to UC-06 Book Ticket.
Alternate / exception flows	2a. A seat is taken by another user at the same moment: the map refreshes and the passenger chooses another seat. 3a. The rule is not met: a warning suggests suitable seats. 4a. The timer reaches zero: the seats are released and the passenger is told.
Postconditions	The selected seats are locked to this passenger for 8 minutes.
Related requirements	FR-12 to FR-15, FR-17, NFR-02, NFR-08

UC-06 · Book Ticket	
Primary actor	Passenger
Secondary actors	Payment Gateway (includes UC-07 Make Payment)
Description	The passenger enters traveller details, pays and receives an e-ticket.
Preconditions	Seats are locked for the passenger (UC-04).
Trigger	The passenger selects Continue.
Main flow	1. The passenger enters name, age and gender for each seat. 2. The passenger chooses boarding and drop points. 3. The system shows the fare summary. 4. The passenger confirms and pays (UC-07). 5. The gateway reports payment success. 6. The system confirms the booking, marks the seats Booked and generates the e-ticket with a QR code. 7. The system sends the ticket by SMS and email (UC-15).
Alternate / exception flows	5a. Payment fails: the passenger may retry while the lock is active. 5b. The lock expired before payment: the seats are released and any captured payment is refunded automatically. 6a. A duplicate confirmation for the same seat is rejected by the database rule.
Postconditions	Booking status is Confirmed; the seats are permanently assigned; the ticket is available.
Related requirements	FR-16, FR-18, FR-19, FR-21 to FR-27, NFR-03, NFR-08, NFR-09

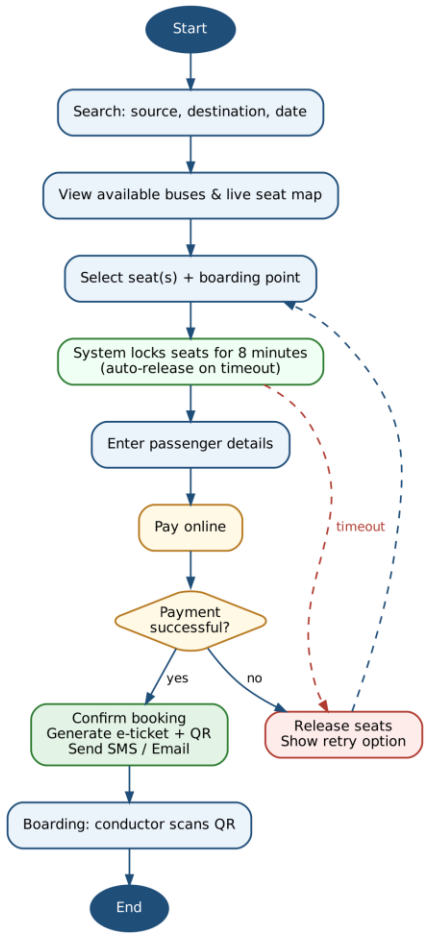


Figure 4: *Booking workflow (UC-04 to UC-07)*

Business rules applied in the booking flow

- A booking may contain up to **6 seats**.
- Seats are held for **8 minutes** from selection.
- A booking is **confirmed only after** the gateway reports payment success.
- If payment fails or the timer ends, seats are **released automatically**.
- The database allows **only one active booking per seat per trip**.
- The e-ticket is sent by SMS and email **within 1 minute**.

> **Why it matters**
> These rules together remove double booking, the main problem found in the scenario.

UC-08 · Join Waitlist	
Primary actor	Passenger (extends UC-06)
Secondary actors	None (notifications through UC-15)
Description	The passenger joins a queue for a full trip and is confirmed automatically when a seat is released.
Preconditions	The trip is Full and the passenger is logged in.
Trigger	The passenger selects Join waitlist.
Main flow	1. The passenger enters the number of seats needed (up to 4). 2. The system records the waitlist position and confirms it by SMS. 3. When a seat is released, the system holds it for the first waitlisted passenger for 15 minutes and notifies them. 4. The passenger pays (UC-07) and the booking is confirmed.
Alternate / exception flows	3a. The passenger does not pay within 15 minutes: the offer passes to the next person. 3b. Departure is less than 2 hours away: the waitlist closes and entries are cancelled.

Postconditions	The waitlist entry is converted to a booking or closed.
Related requirements	FR-20, FR-31

UC-10 · Cancel Ticket	
Primary actor	Passenger
Secondary actors	Payment Gateway (includes UC-11)
Description	The passenger cancels a confirmed ticket and receives a refund based on the refund slabs.
Preconditions	The booking is Confirmed and the trip has not departed.
Trigger	The passenger selects Cancel on a booking.
Main flow	1. The system shows the booking and the refund amount for the current time slab. 2. The passenger confirms the cancellation. 3. The system marks the booking Cancelled and releases the seats. 4. The system calculates the refund and requests it from the gateway (UC-11). 5. The system notifies the passenger and shows the refund status.
Alternate / exception flows	1a. Departure time has passed: cancellation is not allowed. 2a. The passenger declines: nothing changes. 4a. The refund request fails: the status is Refund pending and the system retries automatically; the admin is alerted after 3 failures.
Postconditions	Seats are available for sale; the refund is recorded with its status.
Related requirements	FR-28 to FR-31, FR-35, FR-40

UC-18 · Schedule Trips and Set Fares	
Primary actor	Admin / Operator
Secondary actors	None
Description	The admin schedules one or many trips for a bus on a route and sets the fares.
Preconditions	The admin is logged in; an active bus and a route exist.
Trigger	The admin opens Schedule trip.
Main flow	1. The admin selects a route and a bus. 2. The admin enters the date and times, or chooses a recurrence (daily or weekly until a date). 3. The admin sets fares by seat type. 4. The system checks that the bus is free and that fares are in the allowed range. 5. The system saves the trips with status Scheduled and shows them in passenger search.
Alternate / exception flows	4a. The bus is already assigned at that time: the system rejects the trip. 4b. Fare is zero or above the allowed limit: the system shows an error. 5a. An existing trip with bookings is edited: the system asks for confirmation and notifies affected passengers.

Postconditions	Trips are available for booking.
Related requirements	FR-36, FR-39, FR-45

UC-22 · Scan QR Ticket	
Primary actor	Conductor
Secondary actors	None
Description	The conductor verifies a passenger's e-ticket by scanning its QR code.
Preconditions	The conductor is logged in, the trip is assigned and the phone camera works.
Trigger	A passenger shows the QR code at boarding.
Main flow	1. The conductor scans the QR code. 2. The system checks the code, the trip and the ticket status. 3. The system shows Valid with the passenger name, seat and boarding point. 4. The system marks the ticket Boarded and updates the manifest.
Alternate / exception flows	2a. Ticket already used: the system shows Already used. 2b. Ticket is for another trip: the system shows Different trip. 2c. Ticket was cancelled: the system shows Cancelled. *. No network: the scan is saved on the phone and synchronised later.
Postconditions	The ticket status is Boarded.
Related requirements	FR-43, FR-44

4.7 Data Requirements

4.7.1 Data Entities

Entity	Description	Key attributes
USER	Passenger, admin or conductor account	user_id, name, email, mobile, password_hash, role, status
BUS	A physical bus and its seat layout	bus_id, reg_number, bus_type, seat_layout, total_seats
ROUTE	Journey between two places	route_id, source, destination, distance_km
BOARDING_POINT	A stop on a route	boarding_point_id, route_id, name, time_offset
TRIP	One scheduled run of a bus on a route	trip_id, route_id, bus_id, departure, arrival, fare, status
BOOKING	A passenger's reservation on a trip	booking_id, user_id, trip_id, status, total_fare, booked_at, qr_code
PASSENGER_SEAT	One seat and traveller in a booking	id, booking_id, seat_no, name, age, gender, boarding_point_id

Entity	Description	Key attributes
PAYMENT	Payment and refund record	payment_id, booking_id, amount, mode, status, refund_amount, refund_status
WAITLIST_ENTRY	A queue place for a full trip	id, user_id, trip_id, seats_needed, position, status
REFUND_SLAB	Time-based refund rule	slab_id, hours_before_min, hours_before_max, refund_percent
NOTIFICATION	Message sent to a user	notification_id, user_id, channel, type, sent_at, status
AUDIT_LOG	Record of admin and payment actions	log_id, actor_id, action, entity, timestamp

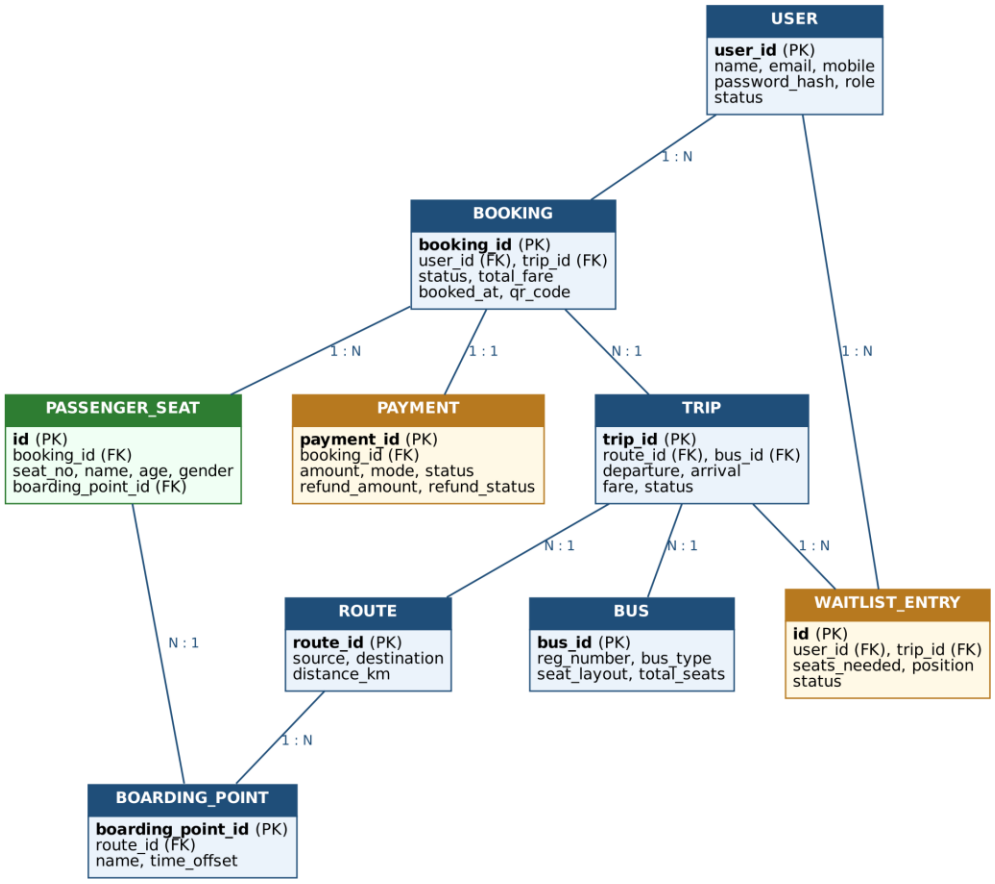


Figure 5: *Entity-relationship diagram of the core data*

4.7.2 Data Rules

ID	Rule
DR-01	A seat of a trip (trip_id and seat_no) can belong to only one active (Locked or Confirmed) booking; this is enforced by the database.
DR-02	Seat locks are kept in a fast cache with an expiry of 8 minutes and are removed automatically when they expire.

ID	Rule
DR-03	Email and mobile number are unique per user; mobile numbers are stored in international format.
DR-04	Passwords and OTPs are stored only as hashes; OTPs expire after 5 minutes.
DR-05	Card details are never stored; only the gateway transaction ID and a masked reference are kept.
DR-06	Bookings, payments and refunds are never deleted; they only change status (Confirmed, Cancelled, Refunded).
DR-07	Booking and payment records are kept for at least 3 years (configurable); personal data is anonymised on a valid deletion request.
DR-08	Times are stored in UTC and shown in Indian Standard Time; amounts are stored in INR with two decimals.
DR-09	Audit log entries are append-only and cannot be edited.
DR-10	Traveller age must be between 0 and 120 and gender must be one of the listed values.

4.8 External Interface Requirements

4.8.1 User Interfaces

Screen	Users	Main elements
Home and search	Guest, Passenger	Source, destination and date fields; popular routes.
Search results	Guest, Passenger	Trip cards with times, fare, seats left; filters and sort.
Seat selection	Passenger	Live seat map with colour legend, countdown timer and fare panel.
Traveller details and payment	Passenger	Traveller forms, boarding point, fare summary, payment options.
My bookings	Passenger	Upcoming and past trips, e-ticket download, cancel button, live tracking.
Admin dashboard and forms	Admin	Sales and occupancy charts; forms for buses, routes, trips, fares and refund slabs.
Conductor scanner	Conductor	Camera view, Valid / Invalid result, trip status buttons.

Interface rules: all screens use one colour scheme and navigation; the seat colour legend and countdown timer are always visible during booking; error messages are written in plain language; every screen works on a 360 px wide phone and meets WCAG 2.1 AA.

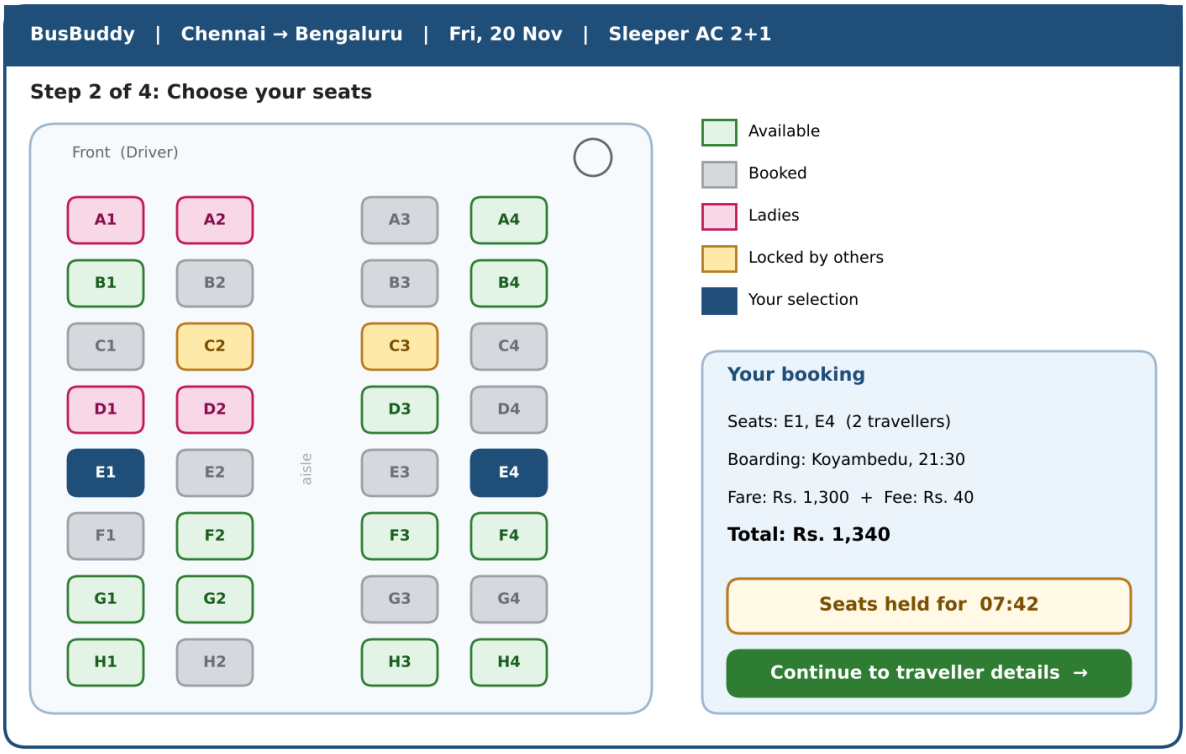


Figure 6: *Wireframe of the seat selection screen (illustrative)*

4.8.2 Hardware Interfaces

- User devices: smartphones, tablets and computers with a web browser. No special hardware is needed by passengers.
- Conductor device: an Android phone with a working camera for QR scanning.
- Optional GPS device or driver app on each bus for live tracking (supports FR-32).
- Servers: cloud virtual machines hosting the application, database and cache.

4.8.3 Software Interfaces

Interface	Purpose	Protocol and data
Payment gateway	Collect payments and issue refunds	HTTPS REST API and signed webhooks; JSON
SMS gateway	Send OTPs, alerts and tickets by SMS	HTTPS REST API; JSON
Email service	Send e-tickets and notices	SMTP or HTTPS API
Maps / GPS service	Bus location and arrival time	HTTPS REST API or SDK; coordinates in JSON
Relational database	Permanent storage of core data	SQL over a secure connection
In-memory cache	Seat locks with expiry	Key-value protocol over a private network

4.8.4 Communication Interfaces

- All client and server traffic uses HTTPS with TLS 1.2 or higher (NFR-04).

- The application exposes a JSON REST API; the seat map is updated by short polling or WebSocket within 2 seconds (NFR-02).
- Webhook calls from the payment gateway are verified by signature before any booking is confirmed.
- API calls are rate-limited per user and per address to prevent misuse.

4.9 System Features

ID	Feature	Description (priority)	Stimulus and response	Requirements
SF-1	**Smart Seat Lock**	Holds chosen seats for 8 minutes during payment (High).	Passenger selects seats, so seats are locked, a timer starts and other users see Locked.	FR-14 to FR-16, NFR-08
SF-2	**Live Seat Map**	Shows real seat status to all users (High).	A seat changes status, so every open map updates within 2 seconds.	FR-12, NFR-02
SF-3	**Women-Safe Seating**	Avoids placing unrelated male passengers beside ladies seats (Medium).	Seat is selected, so the rule is checked and a warning is shown if broken.	FR-17
SF-4	**Online Payment and QR e-Ticket**	Pays online and issues a verifiable ticket (High).	Payment succeeds, so the booking is confirmed and the ticket is sent.	FR-22 to FR-27
SF-5	**Slab-Based Instant Refund**	Calculates refund from time slabs (High).	Passenger cancels, so the refund is shown and initiated automatically.	FR-28 to FR-31, FR-40
SF-6	**Waitlist Auto-Promotion**	Fills released seats from a queue (Low).	A seat is released, so the first waitlisted passenger is offered it.	FR-20
SF-7	**Live Tracking and Delay Alerts**	Shows bus location and warns about delays (Low).	A trip is late, so a delay alert is sent and the ETA is updated.	FR-32, FR-33
SF-8	**Admin Analytics Dashboard**	Shows sales, occupancy and cancellations (Medium).	Admin opens the dashboard, so charts load and can be exported.	FR-41, FR-42

5. Identify Assumptions and Constraints

5.1 Assumptions

Assumptions are things believed to be true while writing this SRS. If one proves false, the requirements may need to change.

ID	Assumption	If it is wrong
A-01	Passengers have a smartphone or computer, internet access and a mobile number for OTP.	Add counter or agent booking.
A-02	Operators enter correct and up-to-date bus, route and fare data.	Wrong trips and fares are shown.
A-03	The payment, SMS and maps services are available and meet their stated uptime.	Fallbacks and retries are needed.
A-04	Every bus has a fixed, known seat layout (seater or sleeper).	Seat-map design must change.
A-05	Conductors have Android phones with a working camera.	Provide scanners or issue devices.
A-06	Buses carry a GPS device or the driver uses a tracking app (needed only for FR-32).	Live tracking is dropped.
A-07	Only domestic routes and the Indian rupee are used in version 1.0.	Currency and tax rules must be added.
A-08	Fares and cancellation rules follow the operator's policy and state regulations.	Refund slabs need revision.
A-09	Passenger gender is entered by the passenger and is not verified.	Add identity checks.
A-10	The project follows Agile Scrum with six two-week sprints.	The release plan must be re-planned.

5.2 Constraints

Type	ID	Constraint	Effect on the project
Technical			
Technical	T-1	The system must use an external payment gateway and follow its API rules and limits.	Design around gateway limits.
Technical	T-2	Standard cloud hosting and an open-source database must be used.	Limits technology choice.
Technical	T-3	Pages must work on slow mobile networks (page weight of 1 MB or less).	Light, optimised screens.
Technical	T-4	Seat locking requires an in-memory cache with expiry.	Extra component to run.

Type	ID	Constraint	Effect on the project
Operational			
Operational	O-1	Operator staff need training; counters keep running during the move to online booking.	Training plan and phased rollout.
Operational	O-2	Planned maintenance is allowed only between 2 AM and 4 AM.	Deployment schedule.
Operational	O-3	Support is available during working hours at launch.	Self-service features are needed.
Economic			
Economic	E-1	The project has a limited budget; open-source tools and free-tier services are preferred.	Careful scope choices.
Economic	E-2	Payment and SMS providers charge per transaction or message.	Running cost per booking.
Economic	E-3	Hosting cost must stay within the budget approved by the operator.	Efficient use of servers.
Legal			
Legal	L-1	The Digital Personal Data Protection Act, 2023 requires consent and deletion on request.	Consent and delete features.
Legal	L-2	Card data must be handled under PCI-DSS, so only a certified gateway handles it.	No card data stored.
Legal	L-3	Consumer-protection and state transport rules apply to fares and refunds.	Fare transparency and refund rules.
Legal	L-4	The Information Technology Act, 2000 governs electronic records and e-tickets.	Records and logs are kept.
Time			
Time	TM-1	The first release must be ready in 12 weeks (six sprints).	Scope is prioritised.
Time	TM-2	All Must-have requirements must be complete by Sprint 3 for the core journey.	Early focus on booking.
Time	TM-3	The academic submission and demonstration date is fixed.	No schedule extension.

6. Validate Requirement Completeness and Consistency

6.1 Stakeholder Coverage

Each stakeholder need was traced to requirements and use cases to check that nothing was missed.

Stakeholder	Key need	Requirements	Use cases	Status
Passengers	Find, book and pay for a seat; cancel; track	FR-07 to FR-35	UC-02 to UC-15	Covered
Bus Operator / Admin	Manage fleet, trips, fares; see reports	FR-36 to FR-42	UC-16 to UC-21	Covered
Conductors / Drivers	Verify tickets; update trips	FR-43, FR-44	UC-22, UC-23	Covered
Payment Gateway Provider	Secure integration	FR-22 to FR-24, FR-30, NFR-05	UC-07, UC-11	Covered
SMS / Email and Maps Providers	Valid API use	FR-26, FR-32 to FR-35	UC-14, UC-15	Covered
Transport Authority	Accurate, auditable records	FR-42, FR-45, DR-06, DR-09	UC-21	Covered
Customer Support Team	Find and fix bookings	FR-27, FR-45	UC-12	Covered
Development Team	Clear, testable requirements	All NFR targets; Section 6.2	All	Covered

Coverage result: all 47 functional requirements are linked to at least one use case, and all 23 use cases are linked to at least one requirement (see Section 3.3).

6.2 Checks for Ambiguity, Redundancy, Inconsistency and Missing Requirements

Check	What was verified	Finding	Action taken
Ambiguity	Words such as fast, secure, quickly, easy	Five statements were vague (for example "search should be fast").	Replaced by measurable targets (3 s, TLS 1.2, 5 min OTP, 3-minute booking).
Redundancy	Duplicate or overlapping requirements	E-ticket delivery appeared under both Payment and Notifications; seat release appeared in booking and cancellation.	Kept once (FR-26) and separated release on timeout (FR-15) from release on cancel (FR-31).
Inconsistency	Conflicting values or rules	Seat lock time was 10 minutes in one place and 8 in another. Session timeout differs for admin and passenger.	Standardised to 8 minutes; the 15 / 30 minute difference is intentional and stated in FR-47.
Missing requirements	Gaps against the scenario and domain	Audit log, session timeout, payment reconciliation, backup and recovery, SMS fallback, accessibility, data deletion and waitlist offer expiry were absent.	Added as FR-45, FR-47, NFR-09, NFR-10, NFR-12, NFR-15, NFR-22 and a rule in UC-08.
Testability	Can each requirement be tested?	All 22 non-functional requirements have a measurable target.	No change needed.
Traceability	Link from need to use case	Every FR maps to a use case; every use case maps to FRs.	Traceability kept in Section 3.3.

6.3 Quality Characteristics of the SRS

Characteristic	Status	Evidence
Correct	Met	Each requirement comes from a stakeholder need or a legal rule (Section 6.1).
Unambiguous	Met	One meaning per statement; measurable targets; a glossary in Section 4.1.3.
Complete	Met	All nine SRS parts are present; actors, use cases, data and interfaces are covered.
Consistent	Met	Values such as the 8-minute lock and 6-seat limit are the same everywhere.
Ranked for importance	Met	Every functional requirement has a MoSCoW priority.
Verifiable	Met	Measurable targets; use case flows and data rules can be turned into test cases.
Modifiable	Met	Unique IDs and one place for each requirement; configurable values noted (NFR-19).
Traceable	Met	Needs, requirements and use cases are linked by IDs.

6.4 Suggested Improvements

- Write acceptance criteria in Given–When–Then form for every Must-have requirement before development starts.
- Validate the screens with real passengers and operators using a clickable prototype, and update the requirements from their feedback.
- Replace the assumed load figures (5,000 users, 100,000 bookings a month) with data from real operators.
- Add a requirement for regional-language support and test it with local users.
- Run a formal requirements review (walkthrough) with all stakeholders, then baseline the SRS and keep a change log.
- Add data-hosting location and data-breach reporting requirements after legal review.

Summary of the Specification

BusBuddy has **47 functional requirements** (27 Must, 15 Should, 5 Could), **22 non-functional requirements**, **8 actors** and **23 use cases**, supported by 12 data entities, 6 software interfaces and 8 system features. All requirements are traceable to stakeholder needs and use cases, and the validation found no open inconsistency.